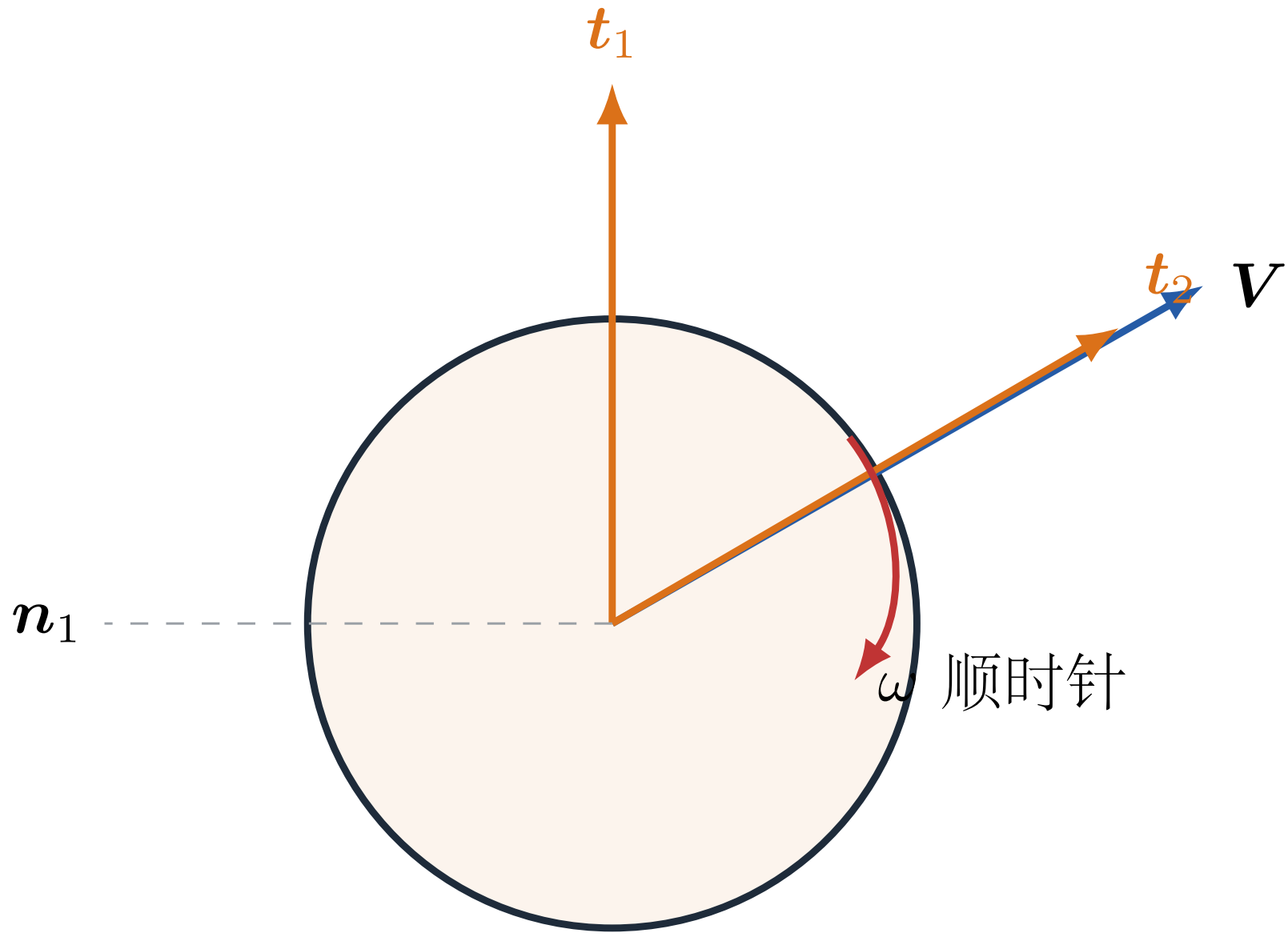

$$\boldsymbol{V} \cdot \boldsymbol{t}_i = v_i - r\omega \equiv u_i$$

$$V = \frac{\sqrt{u_1^2 + u_2^2 - 2u_1 u_2 \cos\alpha}}{\sin\alpha}$$

$$\omega_1 = -\frac{\boldsymbol{V} \cdot \boldsymbol{n}_1}{l_1}$$